1 A student investigates stationary waves with an elastic cord of circular cross-section attached to a load, as shown in Fig. 1.1.

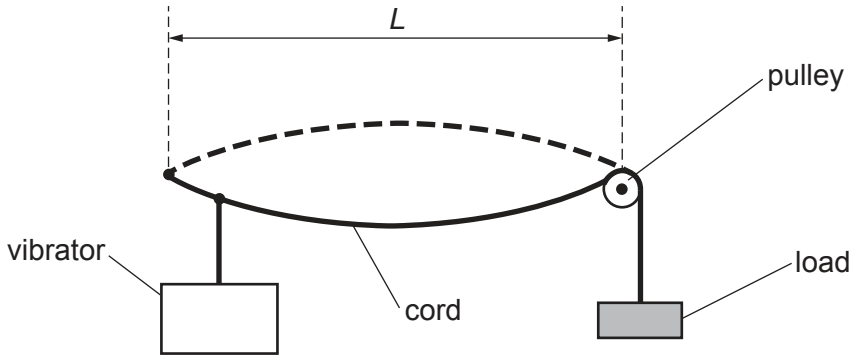

Fig. 1.1

When the frequency of the vibrator is f, the cord vibrates with the stationary wave pattern shown. The student investigates how f varies with the cross-sectional area A of the cord.

It is suggested that the relationship between f and A is

$$f = \frac{1}{2L}\sqrt{\frac{M}{kA}}$$

where L is the distance between the two nodes, M is the mass of the load and k is a constant.

Design a laboratory experiment to test the relationship between f and A.
Explain how your results could be used to determine a value for k.

You should draw a diagram, on page 3, showing the arrangement of your equipment.
In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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